

Out of Syllabus Problems (2005–2014)

Category: Legacy & Discontinued Topics • Legacy Problem Collection

TOTAL QUESTIONS

37

TOTAL MARKS

190 M

TEST DURATION

111 Min

PAPER PATTERN

Classic Paper Pattern (2

Section / Type		Questions	Marking Scheme	Negative Marking	Format / Details
Multiple Choice (MCQ)		37	+1 / +2 / +6 Marks	-1/3 / -2/3 / -2 Marks	4 Options (1 Correct)

IMPORTANT INSTRUCTIONS FOR CANDIDATES

- Historical & Discontinued Topics:** This collection contains authentic official IIT JAM Mathematics questions from historical topics (such as Vector Calculus including Green's, Stokes', and Gauss Divergence Theorems, Mechanics, Statics, and Dynamics) that were tested in earlier years (2005–2021) but have been phased out of the modern JAM MA syllabus.
- Problem-Solving & Entrance Value:** These problems remain exceptionally valuable for competitive mathematical examinations (TIFR, GATE MA, CSIR-NET, NBHM, and CMI/ISI admissions) and deepening advanced analytical intuition.
- Section A (MCQ):** Four options (A), (B), (C), (D) where **only one** is correct. Incorrect attempts incur standard negative marking penalty.
- Section B (MSQ):** Four options (A), (B), (C), (D) where **one or more** options may be correct. Full marks awarded only if all correct choices and zero incorrect choices are chosen. No partial or negative marking.
- Section C (NAT):** Numerical answers are real numbers within the official accepted interval. No negative marking.
- Master Answer Key:** Complete official answers and acceptance bounds are provided on the final page(s) of this document.
- Online Interactive Simulator:** Practice full papers and topic test series at vaibhavgeometer.github.io.

IIT JAM Mathematics Mock Test Series • Curated by Vaibhav Geometer

Question 1 (JAM 2014 • Q35 • ID: JAM_2014_Q35)

MCQ

+2 M

Neg: -0.67

Q.35 Let $S = \{x \in \mathbb{R} : x^6 - x^5 \leq 100\}$ and $T = \{x^2 - 2x : x \in (0, \infty)\}$. The set $S \cap T$ is

- (A) closed and bounded in $\mathbb{R}$
- (B) closed but NOT bounded in $\mathbb{R}$
- (C) bounded but NOT closed in $\mathbb{R}$
- (D) neither closed nor bounded in $\mathbb{R}$

Question 2 (JAM 2014 • Q32 • ID: JAM_2014_Q32)

MCQ

+2 M

Neg: -0.67

Q.32 Let S be the oriented surface $x^2 + y^2 + z^2 = 1$ with the unit normal $\mathbf{n}$ pointing outward. For the vector field $\mathbf{F}(x, y, z) = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$, the value of $\iint_S \mathbf{F} \cdot \mathbf{n} \, dS$ is

- (A) $\frac{\pi}{3}$
- (B) 2π
- (C) $\frac{4\pi}{3}$
- (D) 4π

Question 3 (JAM 2014 • Q29 • ID: JAM_2014_Q29)

MCQ

+2 M

Neg: -0.67

Q.29 Let C be the boundary of the region $R = \{(x, y) \in \mathbb{R}^2 : -1 \leq y \leq 1, 0 \leq x \leq 1 - y^2\}$ oriented in the counterclockwise direction. Then the value of $\oint_C y \, dx + 2x \, dy$ is

- (A) $-\frac{4}{3}$
- (B) $-\frac{2}{3}$
- (C) $\frac{2}{3}$
- (D) $\frac{4}{3}$

Question 4 (JAM 2014 • Q28 • ID: JAM_2014_Q28)

MCQ

+2 M

Neg: -0.67

Q.28 If C is a smooth curve in $\mathbb{R}^3$ from $(-1, 0, 1)$ to $(1, 1, -1)$, then the value of $\int_C (2xy + z^2) \, dx + (x^2 + z) \, dy + (y + 2xz) \, dz$ is

- (A) 0
- (B) 1
- (C) 2
- (D) 3

Question 5 (JAM 2014 • Q21 • ID: JAM_2014_Q21)

MCQ

+2 M

Neg: -0.67

Q.21 The set of all limit points of the set $\left\{ \frac{2}{x+1} : x \in (-1, 1) \right\}$ in $\mathbb{R}$ is

- (A) $[1, \infty)$ (B) $(1, \infty)$ (C) $[-1, 1]$ (D) $[-1, \infty)$

Question 6 (JAM 2014 • Q20 • ID: JAM_2014_Q20)

MCQ

+2 M

Neg: -0.67

Q.20 The set $\left\{ \frac{x^2}{1+x^2} : x \in \mathbb{R} \right\}$ is

- (A) connected but NOT compact in $\mathbb{R}$ (B) compact but NOT connected in $\mathbb{R}$
 (C) compact and connected in $\mathbb{R}$ (D) neither compact nor connected in $\mathbb{R}$

Question 7 (JAM 2013 • Q3 • ID: JAM_2013_Q3)

MCQ

+2 M

Neg: -0.67

Q.3 Let A and B be subsets of $\mathbb{R}$. Which of the following is **NOT** necessarily true?

- (A) $(A \cap B)^o \subseteq A^o \cap B^o$ (B) $A^o \cup B^o \subseteq (A \cup B)^o$
 (C) $\overline{A \cup B} \subseteq \overline{A} \cup \overline{B}$ (D) $\overline{A \cap B} \subseteq \overline{A} \cap \overline{B}$

Question 8 (JAM 2013 • Q2 • ID: JAM_2013_Q2)

MCQ

+2 M

Neg: -0.67

Q.2 The value of n for which the divergence of the function $\vec{F} = \frac{\vec{r}}{|\vec{r}|^n}$, $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$, $|\vec{r}| \neq 0$, vanishes is

- (A) 1 (B) -1 (C) 3 (D) -3

Question 9 (JAM 2012 • Q15 • ID: JAM_2012_Q15)

MCQ

+6 M

Neg: -2.0

Q.15 If C is a smooth curve in $\mathbb{R}^3$ from $(0,0,0)$ to $(2,1,-1)$, then the value of

$$\int_C (2xy + z)dx + (z + x^2)dy + (x + y)dz$$

is

(A) -1

(B) 0

(C) 1

(D) 2

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Question 10 (JAM 2012 • Q14 • ID: JAM_2012_Q14)

MCQ

+6 M

Neg: -2.0

Q.14 If $Y = \left\{ \frac{x}{1+|x|} \mid x \in \mathbb{R} \right\}$, then the set of all limit points of Y is

(A) $(-1,1)$

(B) $(-1,1]$

(C) $[0,1]$

(D) $[-1,1]$

Question 11 (JAM 2012 • Q9 • ID: JAM_2012_Q9)

MCQ

+6 M

Neg: -2.0

Q.9 Which one of the following is TRUE ?

(A) The characteristic of the ring $6\mathbb{Z}$ is 6

(B) The ring $6\mathbb{Z}$ has a zero divisor

(C) The characteristic of the ring $(\mathbb{Z}/6\mathbb{Z}) \times 6\mathbb{Z}$ is zero

(D) The ring $6\mathbb{Z} \times 6\mathbb{Z}$ is an integral domain

Question 12 (JAM 2011 • Q15 • ID: JAM_2011_Q15)

MCQ

+6 M

Neg: -2.0

Q.15 Consider the following subsets of $\mathbb{R}$:

$$E = \left\{ \frac{n}{n+1} : n \in \mathbb{N} \right\}, \quad F = \left\{ \frac{1}{1-x} : 0 \leq x < 1 \right\}.$$

Then

- (A) Both E and F are closed
 (B) E is closed and F is NOT closed
 (C) E is NOT closed and F is closed
 (D) Neither E nor F is closed

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Question 13 (JAM 2011 • Q8 • ID: JAM_2011_Q8)

MCQ

+6 M

Neg: -2.0

Q.8 Consider the vector field $\vec{F} = (ax + y + a)\hat{i} + \hat{j} - (x + y)\hat{k}$, where a is a constant. If $\vec{F} \cdot \text{curl } \vec{F} = 0$, then the value of a is

- (A) -1 (B) 0 (C) 1 (D) $\frac{3}{2}$

Question 14 (JAM 2011 • Q7 • ID: JAM_2011_Q7)

MCQ

+6 M

Neg: -2.0

Q.7 Let $\vec{F} = ay\hat{i} + z\hat{j} + x\hat{k}$ and C be the positively oriented closed curve given by $x^2 + y^2 = 1, z = 0$. If $\oint_C \vec{F} \cdot d\vec{r} = \pi$, then the value of a is

- (A) -1 (B) 0 (C) $\frac{1}{2}$ (D) 1

Question 15 (JAM 2010 • Q8 • ID: JAM_2010_Q8)

MCQ

+6 M

Neg: -2.0

Q.8 Let $\mathbf{R}[X]$ be the ring of real polynomials in the variable X . The number of ideals in the quotient ring $\mathbf{R}[X]/(X^2 - 3X + 2)$ is

- (A) 2 (B) 3 (C) 4 (D) 6

Question 16 (JAM 2010 • Q7 • ID: JAM_2010_Q7)

MCQ

+6 M

Neg: -2.0

Q.7 Let $\vec{F} = xy\hat{i} + y\hat{j} - yz\hat{k}$ denote the force field on a particle traversing the path L from $(0, 0, 0)$ to $(1, 1, 1)$ along the curve of intersection of the cylinder $y = x^2$ and the plane $z = x$. The work done by $\vec{F}$ is

- (A) 0 (B) $\frac{1}{4}$ (C) $\frac{1}{2}$ (D) 1

Question 17 (JAM 2010 • Q6 • ID: JAM_2010_Q6)

MCQ

+6 M

Neg: -2.0

Q.6 Let $\vec{F} = 2xyz e^{x^2}\hat{i} + z e^{x^2}\hat{j} + y e^{x^2}\hat{k}$ be the gradient of a scalar function. The value of $\int_L \vec{F} \cdot d\vec{r}$ along the oriented path L from $(0, 0, 0)$ to $(1, 0, 2)$ and then to $(1, 1, 2)$ is

- (A) 0 (B) $2e$ (C) e (D) e^2

Question 18 (JAM 2010 • Q4 • ID: JAM_2010_Q4)

MCQ

+6 M

Neg: -2.0

Q.4 Let S be an infinite subset of $\mathbf{R}$ such that $S \cap \mathbf{Q} = \emptyset$. Which of the following statements is true?

- (A) S must have a limit point which belongs to $\mathbf{Q}$
 (B) S must have a limit point which belongs to $\mathbf{R} \setminus \mathbf{Q}$
 (C) S cannot be a closed set in $\mathbf{R}$
 (D) $\mathbf{R} \setminus S$ must have a limit point which belongs to S

Question 19 (JAM 2009 • Q15 • ID: JAM_2009_Q15)

MCQ

+6 M

Neg: -2.0

Q.15 The value of c for which there exists a twice differentiable vector field $\vec{F}$ with $\text{curl } \vec{F} = 2x\hat{i} - 7y\hat{j} + cz\hat{k}$ is

- (A) 0 (B) 2 (C) 5 (D) 7

Question 20 (JAM 2009 • Q13 • ID: JAM_2009_Q13)

MCQ

+6 M

Neg: -2.0

Q.13 Suppose $\vec{V} = p(x, y)\hat{i} + q(x, y)\hat{j}$ is a continuously differentiable vector field defined in a domain D in $\mathbb{R}^2$. Which one of the following statements is NOT equivalent to the remaining ones?

- (A) There exists a function $\phi(x, y)$ such that $\frac{\partial \phi}{\partial x} = p(x, y)$ and $\frac{\partial \phi}{\partial y} = q(x, y)$ for all $(x, y) \in D$
 (B) $\frac{\partial q}{\partial x} = \frac{\partial p}{\partial y}$ holds at all points of D
 (C) $\oint_C \vec{V} \cdot d\vec{r} = 0$ for every piecewise smooth closed curve C in D
 (D) The differential $pdx + qdy$ is exact in D

Question 21 (JAM 2009 • Q2 • ID: JAM_2009_Q2)

MCQ

+6 M

Neg: -2.0

Q.2 Let R be the ring of all functions from $\mathbb{R}$ to $\mathbb{R}$ under point-wise addition and multiplication. Let $I = \{f: \mathbb{R} \rightarrow \mathbb{R} \mid f \text{ is a bounded function}\}$, $J = \{f: \mathbb{R} \rightarrow \mathbb{R} \mid f(3) = 0\}$. Then

- (A) J is an ideal of R but I is not an ideal of R
 (B) I is an ideal of R but J is not an ideal of R
 (C) both I and J are ideals of R
 (D) neither I nor J is an ideal of R

Question 22 (JAM 2008 • Q10 • ID: JAM_2008_Q10)

MCQ

+6 M

Neg: -2.0

- Q.10 The set $R = \{f \mid f \text{ is a function from } \mathbb{Z} \text{ to } \mathbb{R}\}$ under the binary operations $+$ and $\cdot$ defined as $(f+g)(n) = f(n) + g(n)$ and $(f \cdot g)(n) = f(n)g(n)$ for all $n \in \mathbb{Z}$ forms a ring. Let $S_1 = \{f \in R \mid f(-n) = f(n) \text{ for all } n \in \mathbb{Z}\}$ and $S_2 = \{f \in R \mid f(0) = 0\}$. Then
- (A) S_1 and S_2 are both ideals in R (B) S_1 is an ideal in R while S_2 is not
(C) S_2 is an ideal in R while S_1 is not (D) neither S_1 nor S_2 is an ideal in R

Question 23 (JAM 2008 • Q6 • ID: JAM_2008_Q6)

MCQ

+6 M

Neg: -2.0

- Q.6 Let $V = \{(x, y, z) \in \mathbb{R}^3 : \frac{1}{4} \leq x^2 + y^2 + z^2 \leq 1\}$ and $\vec{F} = \frac{x\hat{i} + y\hat{j} + z\hat{k}}{(x^2 + y^2 + z^2)^2}$ for $(x, y, z) \in V$. Let $\hat{n}$ denote the outward unit normal vector to the boundary of V and S denote the part $\{(x, y, z) \in \mathbb{R}^3 : x^2 + y^2 + z^2 = \frac{1}{4}\}$ of the boundary of V . Then $\iint_S \vec{F} \cdot \hat{n} \, dS$ is equal to
- (A) -8π (B) -4π (C) 4π (D) 8π

Question 24 (JAM 2008 • Q5 • ID: JAM_2008_Q5)

MCQ

+6 M

Neg: -2.0

- Q.5 The set of all boundary points of $\mathbb{Q}$ in $\mathbb{R}$ is
- (A) $\mathbb{R}$ (B) $\mathbb{R} \setminus \mathbb{Q}$ (C) $\mathbb{Q}$ (D) $\emptyset$

Question 25 (JAM 2008 • Q4 • ID: JAM_2008_Q4)

MCQ

+6 M

Neg: -2.0

- Q.4 The work done by the force $\vec{F} = 4y\hat{i} - 3xy\hat{j} + z^2\hat{k}$ in moving a particle over the circular path $x^2 + y^2 = 1, z = 0$ from $(1, 0, 0)$ to $(0, 1, 0)$ is
- (A) $\pi + 1$ (B) $\pi - 1$ (C) $-\pi + 1$ (D) $-\pi - 1$

Question 26 (JAM 2007 • Q15 • ID: JAM_2007_Q15)

MCQ

+6 M

Neg: -2.0

15. Let C denote the boundary of the semi-circular disk $D = \{(x, y) \in \mathbf{R}^2 : x^2 + y^2 \leq 1, y \geq 0\}$ and let $\varphi(x, y) = x^2 + y$ for $(x, y) \in D$. If $\hat{n}$ is the outward unit normal to C , then the integral $\oint_C (\vec{\nabla} \varphi) \cdot \hat{n} ds$, evaluated counter-clockwise over C , is equal to

- (A) 0
- (B) $\pi - 2$
- (C) π
- (D) $\pi + 2$

Question 27 (JAM 2007 • Q13 • ID: JAM_2007_Q13)

MCQ

+6 M

Neg: -2.0

13. Let $\vec{u} = (ae^x \sin y - 4x)\hat{i} + (2y + e^x \cos y)\hat{j} + az\hat{k}$, where a is a constant. If the line integral $\oint_C \vec{u} \cdot d\vec{r}$ over every closed curve C is zero, then a is equal to

- (A) -2
- (B) -1
- (C) 0
- (D) 1

Question 28 (JAM 2007 • Q4 • ID: JAM_2007_Q4)

MCQ

+6 M

Neg: -2.0

4. Let R be the ring of polynomials over $\mathbf{Z}_2$ and let I be the ideal of R generated by the polynomial $x^3 + x + 1$. Then the number of elements in the quotient ring R/I is
- (A) 2
 - (B) 4
 - (C) 8
 - (D) 16

Question 29 (JAM 2006 • Q14 • ID: JAM_2006_Q14)

MCQ

+6 M

Neg: -2.0

14. Let R be the ring of all 2×2 matrices with integer entries. Which of the following subsets of R is an integral domain?

- (A) $\left\{ \begin{pmatrix} 0 & x \\ y & 0 \end{pmatrix} : x, y \in \mathbf{Z} \right\}$
- (B) $\left\{ \begin{pmatrix} x & 0 \\ 0 & y \end{pmatrix} : x, y \in \mathbf{Z} \right\}$
- (C) $\left\{ \begin{pmatrix} x & 0 \\ 0 & x \end{pmatrix} : x \in \mathbf{Z} \right\}$
- (D) $\left\{ \begin{pmatrix} x & y \\ y & z \end{pmatrix} : x, y, z \in \mathbf{Z} \right\}$

Question 30 (JAM 2006 • Q9 • ID: JAM_2006_Q9)

MCQ

+6 M

Neg: -2.0

9. Let S be a closed surface for which $\iint_S \vec{r} \cdot \hat{n} d\sigma = 1$. Then the volume enclosed by the surface is

- (A) 1
- (B) $1/3$
- (C) $2/3$
- (D) 3

Question 31 (JAM 2006 • Q7 • ID: JAM_2006_Q7)

MCQ

+6 M

Neg: -2.0

7. Let r be the distance of a point $P(x, y, z)$ from the origin O . Then ∇r is a vector

- (A) orthogonal to $\vec{OP}$
- (B) normal to the level surface of r at P
- (C) normal to the surface of revolution generated by OP about x-axis
- (D) normal to the surface of revolution generated by OP about y-axis

Question 32 (JAM 2006 • Q6 • ID: JAM_2006_Q6)

MCQ

+6 M

Neg: -2.0

6. Let C be the circle $x^2 + y^2 = 1$ taken in the anti-clockwise sense. Then the value of the integral

$$\int_C \left[(2xy^3 + y)dx + (3x^2y^2 + 2x)dy \right]$$

equals

- (A) 1
- (B) $\pi/2$
- (C) π
- (D) 0

Question 33 (JAM 2006 • Q4 • ID: JAM_2006_Q4)

MCQ

+6 M

Neg: -2.0

4. Let G be the set of all irrational numbers. The interior and the closure of G are denoted by G^0 and $\overline{G}$, respectively. Then

- (A) $G^0 = \emptyset$, $\overline{G} = G$
- (B) $G^0 = \mathbb{I}$, $\overline{G} = \mathbb{I}$
- (C) $G^0 = \emptyset$, $\overline{G} = \mathbb{I}$
- (D) $G^0 = G$, $\overline{G} = \mathbb{I}$

Question 34 (JAM 2005 • Q13 • ID: JAM_2005_Q13)

MCQ

+6 M

Neg: -2.0

13. Let $f: \mathbf{R} \longrightarrow \mathbf{R}$ be defined by $f(t) = t^2$ and let U be any non-empty open subset of $\mathbf{R}$. Then

- (A) $f(U)$ is open
- (B) $f^{-1}(U)$ is open
- (C) $f(U)$ is closed
- (D) $f^{-1}(U)$ is closed

Question 35 (JAM 2005 • Q9 • ID: JAM_2005_Q9)

MCQ

+6 M

Neg: -2.0

9. A ring R has maximal ideals

- (A) if R is infinite
- (B) if R is finite
- (C) if R is finite with at least 2 elements
- (D) only if R is finite

Question 36 (JAM 2005 • Q5 • ID: JAM_2005_Q5)

MCQ

+6 M

Neg: -2.0

5. Let $\vec{F} = x\hat{i} + 2y\hat{j} + 3z\hat{k}$, S be the surface of the sphere $x^2 + y^2 + z^2 = 1$ and $\hat{n}$ be the inward unit normal vector to S . Then $\iint_S \vec{F} \cdot \hat{n} dS$ is equal to

- (A) 4π
- (B) -4π
- (C) 8π
- (D) -8π

Question 37 (JAM 2005 • Q4 • ID: JAM_2005_Q4)

MCQ

+6 M

Neg: -2.0

4. Let $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$. If $\phi(x, y, z)$ is a solution of the Laplace equation then the vector field $\left(\vec{\nabla}\phi + \vec{r}\right)$ is

- (A) neither solenoidal nor irrotational
- (B) solenoidal but not irrotational
- (C) both solenoidal and irrotational
- (D) irrotational but not solenoidal

OFFICIAL MASTER ANSWER KEY & EVALUATION RUBRIC

Out of Syllabus Problems (2005–2014) • All Official Answers & Acceptance Ranges

Q. No.	Type	Marks	Official Key
1	MCQ	+2	A
2	MCQ	+2	D
3	MCQ	+2	D
4	MCQ	+2	C
5	MCQ	+2	A
6	MCQ	+2	A
7	MCQ	+2	D
8	MCQ	+2	C
9	MCQ	+6	C
10	MCQ	+6	D
11	MCQ	+6	C
12	MCQ	+6	C
13	MCQ	+6	D

Q. No.	Type	Marks	Official Key
14	MCQ	+6	A
15	MCQ	+6	C
16	MCQ	+6	C
17	MCQ	+6	B
18	MCQ	+6	D
19	MCQ	+6	C
20	MCQ	+6	B
21	MCQ	+6	A
22	MCQ	+6	C
23	MCQ	+6	A
24	MCQ	+6	A
25	MCQ	+6	D
26	MCQ	+6	C

Q. No.	Type	Marks	Official Key
27	MCQ	+6	D
28	MCQ	+6	C
29	MCQ	+6	C
30	MCQ	+6	B
31	MCQ	+6	B
32	MCQ	+6	C
33	MCQ	+6	C
34	MCQ	+6	B
35	MCQ	+6	C
36	MCQ	+6	D
37	MCQ	+6	D

SCORING METHODOLOGY & EVALUATION GUIDELINES

- **Multiple Choice Questions (MCQ):** Full marks for correct single choice. Negative marks deducted for incorrect attempts. Unattempted questions award zero.
- **Multiple Select Questions (MSQ):** Full marks if and only if all correct choices are marked and no incorrect choice is selected. Zero marks for partial selection or any incorrect choice. No negative marking.
- **Numerical Answer Type (NAT):** Any numerical value falling strictly within the specified official range receives full credit. No negative marking.
- **Marks to All (MTA):** Questions officially designated as MTA award full marks to all candidates regardless of attempt.